Thm. Let $\{a_n\}$ be a sequence of real numbers, and let $S_k \overset{\text{def}}{=} a_1 + \cdots + a_k$ $(k \in \mathbb{N})$.
If $\lim_{n\to\infty} a_n = A \in \mathbb{R}$, then $\lim_{n\to\infty} \frac{S_n}{n} = A$.

Proof. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow |a_n - A| < \varepsilon$
Meanwhile, observe that

$$\left|\frac{S_n}{n} - A\right| = \frac{1}{n}\cdot\left|a_1 + \cdots + a_n - nA\right| = \frac{1}{n}\cdot\left|a_1 - A + a_2 - A + \cdots + a_n - A\right|$$

Now, for all $n \geq N$,

$$\begin{aligned}
\left|\frac{S_n}{n} - A\right| &= \frac{1}{n}\cdot\left|a_1 - A + \cdots + a_N - A + a_{N+1} - A + \cdots + a_n - A\right| \\
&\leq \frac{1}{n}\cdot\left(|a_1 - A| + |a_2 - A| + \cdots + |a_n - A|\right) \\
&= \frac{1}{n}\cdot\left(\sum_{i=1}^{N}|a_i - A|\right) + \frac{1}{n}\cdot\left(\sum_{i=N+1}^{n}|a_i - A|\right) \\
&\leq \frac{1}{n}\cdot\left(\sum_{i=1}^{N}|a_i - A|\right) + \frac{1}{n}\cdot(n-N)\cdot\varepsilon \\
&\leq \frac{1}{n}\cdot\left(\sum_{i=1}^{N}|a_i - A|\right) + \varepsilon
\end{aligned}$$

Since N was fixed, $\limsup_{n\to\infty}\left|\frac{S_n}{n} - A\right| \leq \limsup_{n\to\infty}\frac{1}{n}\left(\sum_{i=1}^{N}|a_i - A|\right) + \varepsilon = \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, $\limsup_{n\to\infty}\left|\frac{S_n}{n} - A\right| = 0$, so $\lim_{n\to\infty}\frac{S_n}{n} = A$. $\blacksquare$

This theorem is called the Cesaro mean,
and in the next page, we deal with the generalization which is called the Stolz-Cesaro theorem.

「Stolz-Cesaro Theorem.

Let $\{a_n\}$ be a sequence of real numbers, and $\{b_n\}$ be a strictly increasing which diverges to $+\infty$.

Then, $\displaystyle \liminf_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} \overset{①}{\leq} \liminf_{n\to\infty} \frac{a_n}{b_n} \overset{②}{\leq} \limsup_{n\to\infty} \frac{a_n}{b_n} \overset{③}{\leq} \limsup_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n}$.

Moreover, if the limit $\displaystyle \lim_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n}$ exists,

then $\displaystyle \lim_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \lim_{n\to\infty} \frac{a_n}{b_n}$

Proof. To justify the inequalities, enough to show that the ①, because replacing $\{a_n\}$ to $\{-a_n\}$ in ①, we obtain ③ directly, and ② is given by definition.

If the first term in the inequality is $-\infty$, then there is nothing to prove.

Assume that the term has finite value.

Suppose that $\alpha \in \mathbb{R}$ satisfies $\displaystyle \alpha < \liminf_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n}$.

Then, there exists $N \in \mathbb{N}$ such that

$$k \geq N \Rightarrow \alpha < \frac{a_{k+1}-a_k}{b_{k+1}-b_k} \Rightarrow \alpha(b_{k+1}-b_k) < a_{k+1}-a_k.$$

For $m \geq N$, by the telescoping,

$$\alpha \cdot [b_m - b_N] = \alpha \cdot \sum_{i=N+1}^{m} (b_i - b_{i-1}) < \sum_{i=N+1}^{m} (a_i - a_{i-1}) = a_m - a_N$$

dividing $b_m - b_N > 0$ to the both side, then

$$\alpha < \frac{a_m - a_N}{b_m - b_N} = \frac{a_m}{b_m - b_N} - \frac{a_N}{b_m - b_N} = \frac{a_m}{b_m} \cdot \frac{1}{1 - b_N/b_m} - \frac{a_N}{b_m - b_N}$$

Now, since b_m diverges to $+\infty$, taking $\liminf_{m\to\infty}$ to the both side, then

$$\alpha \leq \liminf_{m\to\infty} \frac{a_m}{b_m}.$$

Consequently,

$$\alpha < \liminf_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} \Rightarrow \alpha \leq \liminf_{n\to\infty} \frac{a_n}{b_n} \text{ implies } \liminf_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} \leq \liminf_{n\to\infty} \frac{a_n}{b_n}.$$

(To clarify, consider the negative). Further, the last assertion is obtained by the sqeeze theorem. ☒

「Corollary.

Let $\{x_n\}$ be a sequence of real numbers.

If $\lim_{n\to\infty} x_n = L$, then $\lim_{n\to\infty} \frac{x_1+x_2+\cdots+x_n}{n} = \lim_{n\to\infty} \frac{1}{n}\cdot\left[\sum_{i=1}^{n} x_i\right] = L$.

Proof. Let $a_n = \sum_{k=1}^{n} x_k$ and $b_n = n$, for each $n \in \mathbb{N}$.

Then clearly $b_n \to \infty$ and strictly increase, So by the Stolz-Cesaro Theorem,

$\lim_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \lim_{n\to\infty} x_{n+1} = L$ implies that $\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{x_1+\cdots+x_n}{n} = L$. ▨

「Corollary.

Let $\{x_n\}$ be a sequence of positive real numbers.

If $\lim_{n\to\infty} x_n = L > 0$, then $\lim_{n\to\infty} \sqrt[n]{x_1x_2\cdots x_n} = \lim_{n\to\infty}\left[\prod_{i=1}^{n} x_i\right]^{\frac{1}{n}} = L$.

Proof. Let $y_n = \ln(x_n)$ for each $n\in\mathbb{N}$, then by the above corollary,

$$\lim_{n\to\infty} \frac{y_1+\cdots+y_n}{n} = \lim_{n\to\infty} \ln\left(\sqrt[n]{x_1\cdots x_n}\right) = \ln L,$$

(Since the log function is continuous, it preserves the convergence) So proved ▨

「Corollary.

Let $\{x_n\}$ be a sequence of real numbers, and $\{y_n\}$ be a positive sequence satisfying

$$\sum_{i=1}^{\infty} y_i = \infty.$$

If $\lim_{n\to\infty} \frac{x_n}{y_n} = L$, then $\lim_{n\to\infty} \frac{x_1+\cdots+x_n}{y_1+\cdots+y_n} = L$.

Proof.

Let $a_n = \sum_{k=1}^{n} x_k$ and $b_n = \sum_{k=1}^{n} y_k$ for each $n\in\mathbb{N}$. Since $\{y_k\}$ is positive and the sum diverges to $+\infty$, therefore it satisfies the condition of Cesaro theorem, with $\lim_{n\to\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \lim_{n\to\infty} \frac{x_{n+1}}{y_{n+1}} = L$, so

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{x_1+\cdots+x_n}{y_1+\cdots+y_n} = L.$$

▨